

Applied Linear Algebra

Linear Functions

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Consider the vector inner product function

$$f(x) = a^T x = a_1 x_1 + \cdots + a_n x_n$$

where a is a fixed n -vector, and x is a variable n -vector.

The function f is called a *linear function* of x .

The vector a is called the *coefficient vector* of the linear function f .

a is also called the *weight vector* of the linear function f .

A linear function satisfies the following two properties:

- *Additivity.* $f(x + y) = f(x) + f(y)$ for all n -vectors x and y .
- *Homogeneity.* $f(\alpha x) = \alpha f(x)$ for all n -vectors x and all scalars α .

The vector inner product function

$$f(x) = a^T x = a_1 x_1 + \cdots + a_n x_n, \quad a : \mathbb{R}^n, \quad x : \mathbb{R}^n$$

is linear.

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

for all n -vectors x and y , and all scalars α and β .

Given the inner product function $f(x) = a^T x$, we can verify that it is linear:

$$\begin{aligned} f(\alpha x + \beta y) &= a^T(\alpha x + \beta y) \\ &= a^T(\alpha x) + a^T(\beta y) \\ &= \alpha(a^T x) + \beta(a^T y) \\ &= \alpha f(x) + \beta f(y) \end{aligned}$$

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Unit 1

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